- Problem Statement ID : PS8:- Rural AgriCredit Intelligence & Trust Scoring Platform
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Problem & SOLUTION

The Core Problem We Are Solving:-

1. Farmers Don’t Have Formal Credit History
2. Dependence on Informal Moneylenders
3. No Data-Based Trust System
4. Fraud & Information Asymmetry

The Solution:-

The Core Problem We Are Solving

India’s small and marginal farmers face structural barriers in accessing fair and timely credit. Despite contributing significantly to agricultural production, their financial credibility is often invisible within formal banking systems.

1. Lack of Formal Credit History

Most small farmers operate outside structured financial systems.
They do not possess sufficient documented credit records, making traditional underwriting models inadequate and exclusionary.

2. Dependence on Informal Moneylenders

Due to limited access to institutional credit, farmers frequently rely on informal lenders who charge high interest rates, increasing financial vulnerability and debt cycles.

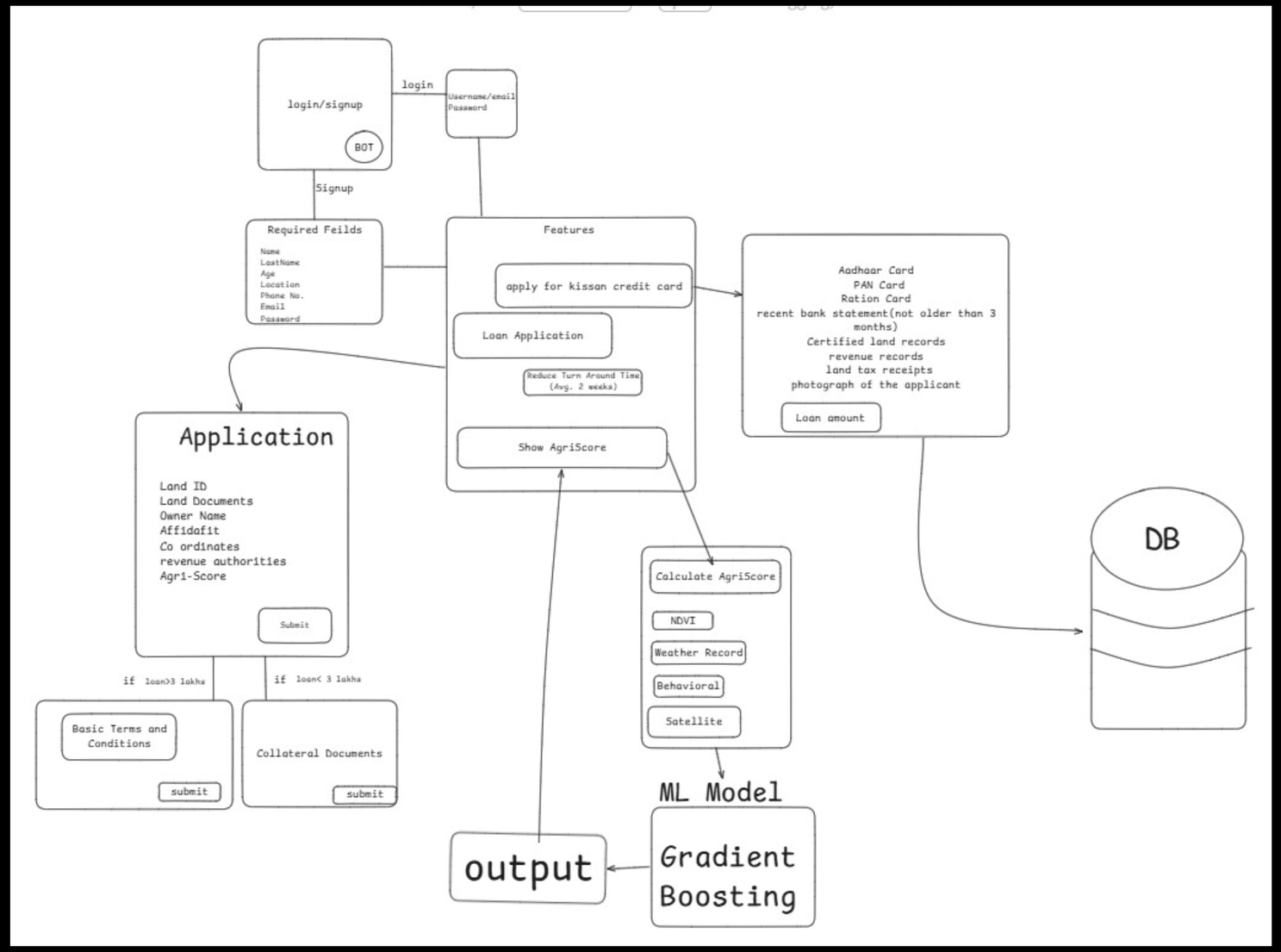
3. Absence of a Data-Driven Agricultural Trust System

Current lending mechanisms are largely collateral-based and manual.
There is no integrated system that evaluates:
Crop performance, Environmental conditions, Yield consistency, Repayment behavior
As a result, credit decisions lack objective, multi-source validation.

4. Fraud & Information Asymmetry

Lenders often have limited visibility into on-ground crop health and productivity.
Manual verification processes create room for:
Misreported yields, Duplicate land claims

Flow of Solution



TECH STACK & APPROACH

Tech Stack used :-

Programming languages :
Python
Javascript

Frontend :
Typescript
React js

Backend :
FastAPI
FireBase

Machine Learning models :
XGBoost

APPROACH:-

This system is a data-driven agricultural credit intelligence layer that evaluates environmental risk, farm productivity, and financial behavior to generate a transparent Agri-Trust Score for lenders.

It aggregates multi-signal inputs such as satellite NDVI, weather patterns, yield history, crop seasonality, repayment behavior, and transaction regularity, then cleans and converts them into structured features.
A XGBoost predicts default probability.

The platform includes explainability features, fraud detection checks (e.g., land duplication, crop mismatch, abnormal yield spikes)

A decision-support dashboard where lenders retain final authority, it supports dynamic re-scoring through seasonal satellite updates, weekly weather risk adjustments, and repayment tracking for continuous monitoring.

UNIQUENESS & INNOVATION FACTOR

Multi-Source Agricultural Intelligence

Performance-Based Lending, Not Collateral-Based

Satellite-Verified Crop Validation

Dynamic, Real-Time Risk Scoring

Explainable & Transparent AI

Fraud & Information Asymmetry Reduction

Plug-In Intelligence Layer

FeASiBility & ChAllenGeS

Feasibility

Data Availability

Technical Viability

Cost-Effective Implementation

Practical Deployment Path

Challenges

Limited Real Default Training Data

Satellite Resolution Limitations

Connectivity Constraints in Rural Areas

Regulatory & Compliance Considerations

Model Bias & Explainability Risks

ReSeARCh & RefeRenCe

NDVI & Remote Sensing in Agriculture

Satellite Data Sources

Agricultural Risk Modeling Studies

Government & Institutional Context

Explainable AI in Finance